

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of M. Tech (EC) (EVD) Examination****May 2018****EC751CMOS VLSI Design-II****Date:05/05/2018,Saturday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below.**

- a. Define channel length modulation. [02]
- b. List out two advantages of differential amplifier over single stage amplifier. [02]
- c. Calculate the transconductance and output impedance for NMOS device. [03]

$$\frac{W}{L} = \frac{50}{0.5}, I_D = 0.5 \text{ mA}, \mu_n C_{ox} = 100 \mu\text{A/V}^2, \lambda = 0.5 \text{ V}^{-1}.$$

- Q – 2.a** Common source stage with degeneration is illustrated in Figure.1. Draw the equivalent small signal model and derive the voltage gain. [04]

Q – 2.b Answer any two questions. [10]

- (i) The common source circuit as depicted in Figure.2 is designed to provide an output voltage swing of 2.2 V with a bias current of 1 mA and a small signal voltage gain of 100. Calculate the dimensions of M_1 and M_2 . $V_{DD} = 3 \text{ V}$, $V_{TH,n} = 0.7 \text{ V}$, $V_{TH,p} = -0.8 \text{ V}$, $\mu_n C_{ox} = 134 \frac{\mu\text{A}}{\text{V}^2}$, $\mu_p C_{ox} = 38 \frac{\mu\text{A}}{\text{V}^2}$, $\lambda_n = 0.1 \text{ V}^{-1}$, $\lambda_p = 0.2 \text{ V}^{-1}$.
- (ii) Explain the MOSFET properties: Sub threshold conduction and Body effect with necessary diagrams.
- (iii) For series-parallel switched capacitor equivalent resistor. Draw the circuit diagram. Derive the expression of equivalent resistance in terms of circuit parameters. And Design a Series-Parallel Switched Capacitor Resistor of $1\text{M}\Omega$ if the clock frequency is 250kHz (if $C_1 = C_2 = C$). Find the value of C.

Q – 3.a Answer the following.

- (i) Draw the complete MOS small signal model at low frequency. [03]
- (ii) Small signal models of NMOS and PMOS devices are identical: True or False. [01]

Q-3. b Answer any two questions. [10]

- (i) Design a switched capacitor summing amplifier using stray insensitive transresistance circuits which gives output voltage during the Φ_2 phase period that is equal to $10 V_1 - 5V_2$, where V_1 and V_2 are held constant during $\Phi_2 - \Phi_1$ period and then resampled for the next period.
- (ii) The cascade of Figure 3 is designed to provide an output swing of 1.9 V with a bias current of 0.5 mA. If $\gamma = 0$ and $\left(\frac{W}{L}\right)_{1-4} = \left(\frac{W}{L}\right)$, $V_{DD} = 3 V$, $V_{TH,n} = 0.7 V$, $V_{TH,p} = -0.8$, $\mu_n C_{ox} = 134 \frac{\mu A}{V^2}$, $\mu_p C_{ox} = 38 \frac{\mu A}{V^2}$ calculate $\frac{W}{L}$.
- (iii) Figure 4 presents common gate amplifier. Derive the small signal gain.

SECTION – II

Q - 4 Answer the questions below.

- a. Define CMRR for differential pair circuit. [02]
- b. Justify: The MOSFET operating in saturation can work as an amplifier. [02]
- c. In which region, MOSFET can be used as a resistor. Derive the equivalent resistor for NMOS device. [02]
- d. The channel charge injection introduces which of the following error in the sampled output voltage. [01]
 - (i) Non-unity Gain
 - (ii) Non-Linearity
 - (iii) Constant Offset voltage
 - (iv) All of these

Q – 5.a Explain cascode stage in detail. Draw its small signal equivalent model. [05]

OR

Q – 5.a Explain source follower configuration and derive the input and output impedance with suitable diagram. [05]

Q – 5.b The differential pair depicted in Figure 5 must provide a gain of 40. Consider a power dissipation of 2mW, $V_{DD} = 1.8 V$, $\lambda_n = 0.1 V^{-1}$, $\lambda_p = 0.2 V^{-1}$, $V_{TH,n} = 0.5V$, $V_{TH,p} = -0.4V$, $\mu_n C_{ox} = 100 \frac{\mu A}{V^2}$, $\mu_p C_{ox} = 50 \mu A/V^2$. Calculate the dimensions of all the transistors. [05]

Q – 5.c What is the effect of channel length modulation on current mirror using MOSFET? How does Cascode current source using MOSFET overcome these limitations. [04]

OR

- Q – 5.c** Explain the diode connected NMOS device and derive the equivalent resistance of a diode connected MOSFET. [04]
- Q – 6.a** Design the circuit of Figure 6. for a voltage gain of 20 and a power budget of 1 mW with $V_{DD} = 1.8\text{ V}$. Assume M_1 operates at the edge of saturation if the input common mode level is 1 V. $\mu_n C_{ox} = 2\mu_p C_{ox} = 100 \frac{\mu\text{A}}{\text{V}^2}$, $V_{TH,n} = 0.5\text{ V}$, $V_{TH,p} = -0.4\text{ V}$, $\lambda_n = 0.5\text{ V}^{-1}$, $\lambda_p = 0.1\text{ V}^{-1}$. [05]
- Q – 6.b** Justify and compare the two stage op amp and gain boosted op amp in context to the following parameters: Gain, Output swing, speed and power dissipation. [05]
- Q – 6.c** Draw the two stage op amp and express the overall voltage gain. [04]

OR

- Q – 6.a** Implement Non-inverting integrator using switched capacitor circuit that is insensitive to parasitic capacitance. Also Derive input-output relationship (transfer function) of this integrator. [05]
- Q-6. b** For Figure 7, Deduce the current in M_4 if all the transistors are in saturation and aspect ratio of M_1 , M_2 , M_3 , M_4 are $(W/L)_1$, $(W/L)_2$, $(W/L)_3$ & $(W/L)_4$ respectively. Suggest a possibility of aspect ratio of MOSFET such that I_{out} is 10 times magnified version of I_{REF} . [05]
- Q – 6.c** Explain common mode feedback for high gain amplification. [04]

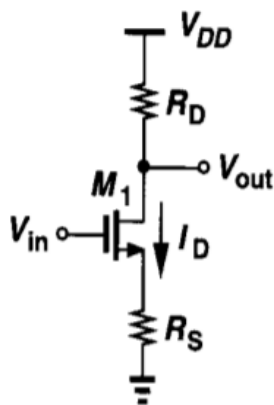


Figure. 1 [Q-2(a)]

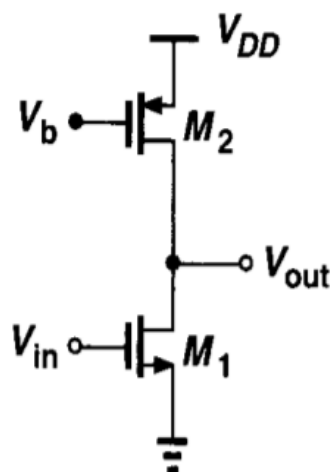


Figure 2 [Q-2b(i)]

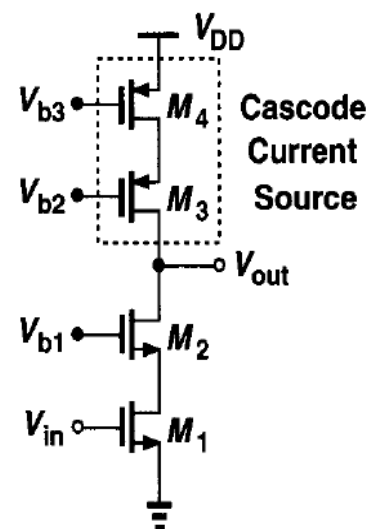


Figure 3 [Q-3b(ii)]

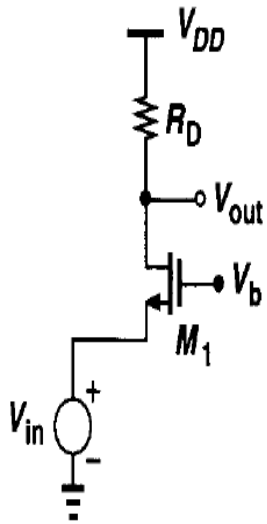


Figure [Q-3b(iii)]

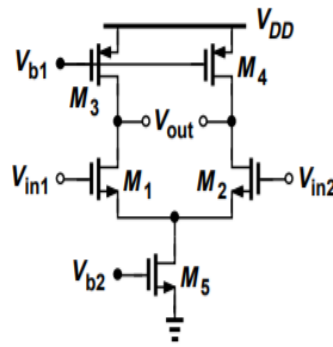


Fig.5[Q-5(b)]

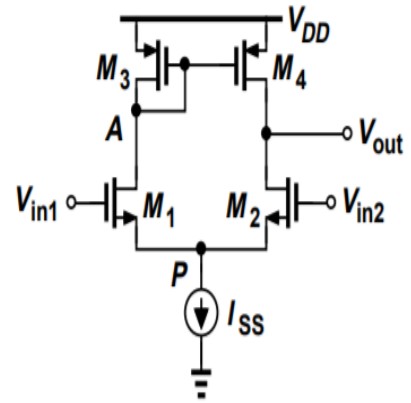


Figure 6 [Q-6(a)]

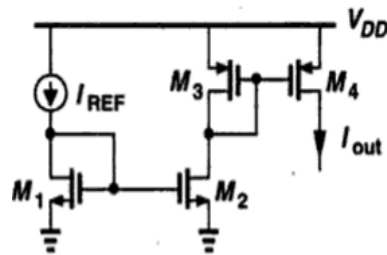


Figure 7.[Q-6(b)]